MELVIN TABITHA OPONDO

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PROFESSIONAL SUMMARY

I am a full-stack Software Developer with hands on experience building responsive web applications using React.js, Django, and Flask. I am proficient in designing RESTful APIs, managing relational and non relational databases, and deploying scalable systems. I work across the full development lifecycle, from architecture and backend logic to frontend interfaces and API integration. I am committed to clean code, continuous learning, and contributing to impactful real world projects.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, C++, HTML, CSS

Frontend: React.js, React Native, TailwindCSS, Bootstrap, Vite, React Hook Form, React Router

Backend & Frameworks: Django, Django REST Framework, Flask, FastAPI

Databases: PostgreSQL, MySQL, MongoDB, SQLite, Redis

Tools & DevOps: Git, GitHub, Docker, Kubernetes (learning), Postman, Ngrok, Vercel, Render, Neon, Oracle Cloud, Netlify, Firebase

APIs & Integrations: RESTful API development, Safaricom Daraja (M-Pesa), Google Maps API, OpenWeather, JWT Authentication

Testing: Pytest, Django Test Framework, React Testing Library, Jest

Other: Technical SEO, Canva, Adobe Photoshop

WORK EXPERIENCE

ACSIS, African Caribbean Sustainability & Investment Summit
Web Developer & Graphic Designer

- I developed and maintained the organisation's website using HTML, CSS, JavaScript, and React.js, improving performance and responsiveness.
- I collaborated with designers to translate branding requirements into intuitive, accessible user interfaces.
- I implemented frontend enhancements that improved site load time and cross-browser compatibility.

PROJECTS

Anga: Weather Platform for Kenya
Live | *Code*

- I designed and built a weather platform for Kenyan users (farmers, outdoor workers, event planners whose decisions depend on accurate local forecasts), engineered to serve live traffic reliably within a strict free tier API quota of 1,000 requests per month. The project is an argument that production grade software can be delivered under the tight resource constraints typical of lean teams and emerging markets.
- I built the backend with Django and Django REST Framework, the frontend with React, and used PostgreSQL for persistence.
- I integrated the OpenWeather API and implemented a usage aware key rotation strategy that spreads live traffic across multiple keys, keeping the service inside quota while real users are being served.
- I verified the upstream provider's live API behaviour and discovered that its documented rate limit headers were absent in practice. I redesigned the caching strategy around observed behaviour rather than documentation, a judgment call that kept the system reliable rather than theoretically correct.
- I built a caching layer with freshness fallbacks and graceful degradation, so the product continues to serve recent data even when the quota is spent. Reliability under constraint, not just under ideal conditions.
- I wrote 222 automated tests across the system (98 backend, 124 frontend) because confidence in a quota constrained system comes from testing, not from live debugging that would itself consume quota.
- I served as sole developer across architecture, API design, data modelling, frontend, and deployment to Vercel.

StudyBuddy: AI-Powered Study Assistant

Live | *Code*

- I built an AI-powered study assistant for students that is built around a single conviction: an AI tutor that invents facts is worse than no tutor at all. Hallucinated definitions and quiz answers presented with false confidence actively harm learners, and the whole system is designed to prevent that outcome rather than apologise for it.
- I implemented a verification layer that checks every AI generated definition and quiz question against the student's own uploaded source notes. If the content cannot be grounded in the source material, it never reaches the learner.
- I designed the interface to say plainly when a response cannot be grounded, rather than fabricate an answer. Honesty is treated as a product feature, not an engineering afterthought, which is what educational AI products should look like.
- I built the backend with Django and Django REST Framework and the frontend with React, backed by PostgreSQL.
- I implemented custom authentication with a three-role permissions model (student, tutor, administrator), and classified errors by type so recovery paths are deterministic rather than guesswork.
- I served as sole developer across the full stack and deployed to Vercel.

CallShield: AI-Powered Scam Call Protection System

- I built an AI-driven call security system that detects and flags potential scam calls in real time, addressing a problem that affects millions of mobile phone users across Africa every day. The product's job is to give a user a signal before they engage with a scam, not after they have lost money.
- I used pattern recognition and data analysis to classify incoming calls, so the model's output can be acted on in the moment rather than after a long investigation.
- I developed the Django backend with RESTful APIs to handle call data processing and machine learning model integration.
- I designed and managed the PostgreSQL database for secure storage of call records and user data.
- I built the frontend using React Native, delivering a responsive and user-friendly mobile experience.
- I served as Project Manager, leading planning, coordinating team tasks, managing timelines, and overseeing testing and deployment readiness. The combination of technical ownership and team leadership reflects the realities of small engineering teams, where the person who ships the code is often also the person who coordinates it.

SmartHunt: Student House Hunting Platform

- I built a student-focused housing platform that connects tenants with verified rental listings and processes B2B payments through the Safaricom Daraja (M-Pesa) API. The project solves a specific real world problem: students arriving in a new city need somewhere safe to live, and the platform integrates with the mobile money infrastructure almost every Kenyan already uses rather than asking them to adopt yet another payment method.
- I developed the backend using Django and Django REST Framework, with FastAPI supporting high-performance services, and designed a MySQL database covering property listings, users, and transactions.
- I integrated the Google Maps API for location-based property discovery, so students can filter listings by proximity to campus rather than scrolling through unrelated options.
- I developed the React frontend, delivering a responsive and intuitive user interface across devices.

UjamaaFlow: Resource Donation Web App

Live | *Code*

- I contributed to a resource donation platform that connects donors with communities in need, structured around a three-role flow for donors, recipients, and administrators. Charitable giving collapses without trust, so the system is designed to make the path from a contribution to its intended recipient visible and auditable rather than taken on faith.
- I integrated donation tracking and payment functionality so every contribution leaves an auditable trail from donor to recipient, meeting the transparency expectations that NGOs and philanthropic organisations have of the software they use.

Master Photography Kenya: Studio Website

Live | *Code*

- I designed and built a marketing website for a Kenyan photography studio, prioritising fast image delivery and a mobile first gallery experience. The constraint was real: the studio's customers browse on phones over variable networks, and a photography portfolio that loads slowly is a photography portfolio nobody looks at.
- Built in vanilla HTML, CSS, and JavaScript with no framework and no build step, demonstrating that polished client work does not require heavy tooling when the fundamentals are used well.
- I deployed the site to Vercel for low latency global distribution.

CSA Africa: Computer Science Academy Website

Live

- I deployed and maintain the live website for Computer Science Academy Africa, owning production hosting, uptime, and the technical SEO fundamentals that keep the organisation discoverable to the students and partners it exists to reach. The role is a demonstration that I can take responsibility for a real external client whose users depend on the site being up and findable, not just a weekend project.

Developer Portfolio

Live

- I designed and built my developer portfolio in vanilla HTML, CSS, and JavaScript without frameworks or a build step. The choice is deliberate: it is a demonstration that I understand the web platform beneath the abstractions I use every day, and that the fundamentals are enough when they are enough.
- The site includes light and dark theming, a live terminal interface, WCAG aware accessibility, and runs identically from any static host or local file system.
- I deployed the site to Vercel.

Weatherfusion: Real-Time Weather Application

Live | *Code*

- I built a Django-powered weather application with a conversational AI assistant that answers natural language weather questions, combining real-time data from an external weather API with a language model to turn structured forecast data into friendlier user experiences.
- The app fetches real-time temperature, humidity, pressure, and forecast data for any city, and the assistant lets a user ask ordinary questions ("will it rain in Nairobi this afternoon?") instead of parsing a dashboard.

Other Projects

- **Pizza Ordering Application.** I built a full-stack ordering system with a JavaScript, HTML, and CSS frontend and a Flask backend. The focus was end to end order flow in real time, so a customer can place, adjust, and check out an order without the state going out of sync between client and server.
- **Health Information System.** I implemented client enrollment into health programmes with profile data exposed only through secured APIs. Health data is only useful if it is safe to share, so the system is designed around controlled access from the beginning rather than as a later retrofit.
- **Voice Powered Calculator.** I built a Python calculator with speech recognition for basic, advanced, and trigonometric operations, exploring voice input as a more accessible interface for users who find typing on small screens difficult.
- **Sentiment Analysis Dashboard.** I built an AI-powered dashboard that turns raw user feedback into visualised sentiment trends in real time, so a product team can see what customers feel without reading every message individually.

EDUCATION & TRAINING

Meru University of Science and Technology

Bachelor of Science in Computer Science

Computer Science Academy (CSA)

Software Engineering

Power Learn Project (PLP)

Software Engineering

Africa Agility Foundation

Frontend Web Development

Workmates Technology

UI/UX Design

ALX, AI Career Essentials Program

AI tools integration and ethical AI applications in industry workflows

Kolanya Girls' School

KCSE | 2019 to 2022

LEADERSHIP & VOLUNTEER EXPERIENCE

Community Outreach Specialist. WISSA, Women in Sustainable STEM Africa

- I supported outreach programmes aimed at empowering women and girls in STEM fields across Africa.

Web & Graphic Designer (Volunteer). ACSIS, African Caribbean Sustainability & Investment Summit

- I designed digital graphics and promotional materials using Adobe Photoshop and Canva.
- I maintained web content to support event visibility, engagement, and organisational branding.

Mentor & STEM Advocate. Kolanya Girls' Secondary School

- I mentored students in STEM career pathways, leadership development, and facilitated coding workshops.

Social Media Manager. Blockchain Community, Meru University Science Innovators Club

- I managed and grew the community's social media presence, creating technical and educational content on blockchain and emerging technologies.
- I supported community engagement, event promotion, and new member onboarding.

SOFT SKILLS

Leadership & Responsibility • Problem Solving & Analytical Thinking • Team Collaboration • Adaptability & Continuous Learning • Effective Communication

LANGUAGES

English, Fluent | Kiswahili, Fluent | French, Intermediate

REFEREES

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